Sports Data Pipeline on AWS · Build Order

Numbered badges show which phase builds each part. Build one phase at a time, one pull request per phase.

Live / streaming flow
Batch flow
Orchestration trigger / quality check
CI/CD deployment flow

ORCHESTRATION

EventBridge Scheduler (7)
polls on a schedule

schedule

Amazon MWAA (managed Airflow) (9)
or Step Functions for lower cost

DAGs trigger the batch backfill, Glue jobs, then dbt build + tests

SOURCES

Live REST APIs (7)
API-Football · NHL API · MLB StatsAPI · balldontlie · OpenF1

Live feeds (8)
SportScore websockets · OpenF1 live · OpenF1 history (replay)

Historical (4)
StatsBomb Open Data · Jolpica / FastF1 · Cricsheet · nflverse · Kaggle

Reference API (4)
TheSportsDB: teams, players (no live data on free plan)

INGESTION

AWS Lambda pollers (7)
call the REST APIs, free tier ~100 calls/day

ECS Fargate task (8)
websocket listeners + OpenF1 replay simulator

Batch backfill (4)
Lambda or Glue job loads StatsBomb, Jolpica, Cricsheet and reference data into S3 raw

STREAMING

Kinesis Data Streams (7)
live event stream

Secrets Manager (7)
API keys for Lambda, Fargate and backfill

DELIVERY

Kinesis Data Firehose (7)
buffers and writes JSON to S3

OpenF1 note (8)
Live data only during sessions; none between sessions or in the off-season (Dec–Mar). Historical (2023+) is free; real-time may need a paid plan. The replay simulator feeds past sessions in time order to mimic a live stream.

RAW STORAGE

Amazon S3 · raw zone (3)
JSON, partitioned by source / date

PROCESSING

AWS Glue jobs (5)
raw JSON → Parquet

CLEAN ZONE

Amazon S3 · clean zone (3)
Parquet tables

Glue Data Catalog (3)
table definitions

schemas

WAREHOUSE

Amazon Athena (6)
low cost · or Redshift Serverless

MODELLING

dbt models (6)
dbt-athena or dbt-redshift
fact_events · dim_team · dim_player · dim_session

OUTPUTS

Amazon QuickSight (12)
dashboards

Notebooks (12)
analysis

DATA QUALITY + MONITORING

Amazon CloudWatch (10)
logs, metrics and alarms for Lambda, Fargate, Kinesis, Firehose, Glue and MWAA

Glue Data Quality (10)
rules on the clean zone

Late or corrected events (10)
keep the latest version per event id; reprocess the affected date partitions

dbt tests (6)
not_null · unique · relationships · freshness

DEPLOYMENT · TERRAFORM + GITHUB CI/CD

GitHub repo (2)
/infra · /ingestion · /dbt · /orchestration · /.github/workflows

GitHub Actions (2)
CI/CD workflows

OIDC → IAM role (1)
assumed by Actions · no stored AWS keys

On pull request (2)
terraform fmt + validate · Checkov / tfsec · plan posted to the PR · pytest for Lambdas · dbt build on dev

On merge to main (11)
Docker image → ECR · package Lambdas · terraform apply to dev · sync DAGs to S3 · run dbt

Terraform (1)
modules in /infra

Terraform state (1)
S3 backend with state locking · separate state for dev and prod

apply

AWS · dev (11)
full pipeline stack

promote

Manual approval (11)

Promote to prod (11)
GitHub Environments approval, then terraform apply to prod

approved

AWS · prod (11)
full pipeline stack

BUILD ORDER · ONE PHASE PER PR

Foundation · Batch path · Live path · Run + harden · Serve

1 · Bootstrap
Terraform state bucket + lock, GitHub OIDC provider, IAM roles (dev, prod)

2 · CI skeleton
PR workflow: fmt / validate, Checkov / tfsec, plan posted to PR, pytest

3 · Storage
S3 raw and clean buckets, Glue Data Catalog database

4 · Batch first
Backfill StatsBomb, Jolpica / FastF1, Cricsheet, TheSportsDB into S3 raw

5 · Processing
Glue jobs raw JSON → Parquet in clean zone; register tables in the Catalog

6 · Warehouse + dbt
Athena workgroup; fact_events, dim_team, dim_player, dim_session + tests

7 · Streaming
Kinesis Streams → Firehose → S3 raw; Lambda pollers on EventBridge; Secrets Manager

8 · Fargate
SportScore websocket listener and OpenF1 replay simulator, image in ECR

9 · Orchestration
DAGs / state machine: backfill → Glue → dbt build + test

10 · Quality + monitoring
Glue Data Quality rules, late / corrected events, CloudWatch alarms

11 · Deploy pipeline
Merge-to-main workflow; prod promotion with GitHub Environments approval

12 · Outputs
QuickSight dashboards, analysis notebooks